Minkyu Jeon

Agentic AI for De Novo Protein Design · Heterogeneous Cryo-EM Reconstruction · Generative Models
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SUMMARY

Ph.D. candidate developing generative and representation-learning methods for scientific inverse problems, focused on (i) **agentic inference-time scaling for de novo protein design**, where LLM agents run binder-design campaigns under a fixed GPU budget by scheduling generative and verification tools (T-REX), and (ii) **heterogeneous cryo-EM reconstruction** of conformational and compositional variability (CryoBench, CryoHype). Research experience at Prescient Design (Genentech) and the Broad Institute.

EDUCATION

Princeton University 2023–present
Ph.D. Candidate in Computer Science (advisor: Prof. Ellen D. Zhong) Princeton, NJ

Korea University 2020–2023
M.S. in Computer Science and Engineering Seoul, Korea

Konkuk University 2014–2020
B.S. in Applied Statistics and Computer Engineering (double major) Seoul, Korea

SELECTED RESEARCH

T-REX: Agentic Campaign Control for De Novo Binder Design 2026–present
First author · bioRxiv 2026 · open-source code

- LLM agents decide when to rescue, explore, or exploit, orchestrating six generation methods (Proteína-Complexa ×4, BindCraft, BoltzGen), ProteinMPNN, and AlphaFold2 evaluation; a deterministic controller validates and schedules every job.
- Highest throughput of structurally distinct in-silico hits on all 7 targets (1,461 total): **2.43**× a tree-search controller and **2.48**× the strongest generator-only baseline per target (geometric mean, matched worker-GPU budgets).

T-REX for Wet-Lab Binder Design Challenges 2026
Anthropic × Adaptyv 2026 · EGFR, TNF-α, GLP1R, TfR1, KRAS G12D pMHC

- Added in-silico proxies for pH-selective binding (Proton-PottsMPNN redesign) and mouse cross-reactivity on EGFR, TNF-α, and TfR1, plus structural-novelty diagnostics.
- Expanded the method pool (BindCraft2, PXDesign, RFdiffusion3, ArcRefine) and ran campaigns with Claude or Qwen as the planning LLM.

Heterogeneous Cryo-EM Reconstruction & Protein Conformation 2023–present

- **CryoBench** (first author, NeurIPS 2024 Datasets & Benchmarks **Spotlight**): benchmark datasets and metrics for conformational and compositional heterogeneity.
- **CryoHype** (co-first, CVPR 2026): transformer hypernetwork that reconstructs 1,000 distinct structures from unlabeled images (fixed-pose).
- **Self-distillation for amortized reconstruction** (co-first, NeurIPS 2025 MLSB Workshop **Oral**).
- **ConforNets** (co-author): latent-space control of alternate conformations in OpenFold3.

EXPERIENCE

Prescient Design, Genentech Nov 2025–Mar 2026
Contractor (part-time) New York, NY
Research Intern Jun 2025–Aug 2025

- Led a project on understanding and improving 3D generative models (diffusion, flow matching) for protein design; studied limitations of voxel- and structure-space models and developed design and evaluation workflows.

Broad Institute of MIT and Harvard Jan 2023–Aug 2023
Associate Computational Biologist (advisor: Dr. Hyunghoon Cho) Cambridge, MA
Visiting Graduate Student Sep 2021–Apr 2022

- Led privacy-preserving retinal image synthesis with GANs (**k-SALSA**, first author, ECCV 2022) and later with diffusion models.

Earlier Research Internships 2018–2020, Seoul, Korea

- **Korea University** (Prof. Hyunwoo J. Kim), 2020: meta-learning for layer-wise optimization (IEEE Access 2021).
- **MakinaRocks**, 2019: anomaly detection with variational autoencoders.
- **KIST** (Dr. Yong Moo Kwon), 2018: situation-aware chatbot.

PUBLICATIONS

[*] equal contribution

AI for Protein Design & Structural Biology

bioRxiv 2026 — **Agentic campaign control for high-throughput de novo binder design**
Minkyu Jeon, Jinyeop Song, Jina Kim, Ellen D. Zhong paper / code

CVPR 2026 — **CryoHype: Reconstructing a thousand cryo-EM structures with transformer-based hypernetworks**
Jeffrey Gu[*], **Minkyu Jeon**[*], Ambri Ma, Serena Yeung-Levy, Ellen D. Zhong paper also NeurIPS 2025 MLSB Workshop (poster)

arXiv 2026 — **ConforNets: Latents-based conformational control in OpenFold3**
Minji Lee, Colin Kalicki, **Minkyu Jeon**, Aymen Qabel, Alisia Fadini, Mohammed AlQuraishi paper

NeurIPS 2025 MLSB Workshop **Oral** — **Separating signal from noise: A self-distillation approach for amortized heterogeneous cryo-EM reconstruction**
Minkyu Jeon[*], Jeffrey Gu[*], Ambri Ma, Serena Yeung-Levy, Vincent Sitzmann, Ellen D. Zhong paper

NeurIPS 2024 D&B **Spotlight** — **CryoBench: Diverse and challenging datasets for the heterogeneity problem in cryo-EM**
Minkyu Jeon, Rishwanth Raghu, Miro Astore, Geoffrey Woollard, Ryan Feathers, Alkin Kaz, Sonya M. Hanson, Pilar Cossio, Ellen D. Zhong paper / project

Machine Learning, Computer Vision & Health

ECCV 2026 Workshop (MUCG) — **CHIMERA: Adaptive cache injection and semantic anchor prompting for zero-shot image morphing with morphing-oriented metrics**
Dahyeon Kye, Jeahun Sung, **Minkyu Jeon**, Jihyong Oh paper / project

arXiv 2025 — **R3eVision: A survey on robust rendering, restoration, and enhancement for 3D low-level vision**
Weeyoung Kwon, Jeahun Sung, **Minkyu Jeon**, Chanho Eom, Jihyong Oh paper

JMIR Med. Inform. 2024 — **A multivariable prediction model for mild cognitive impairment and dementia: Algorithm development and validation**
Sarah Soyeon Oh, Bada Kang, Dahye Hong, Jennifer Ivy Kim, Hyewon Jeong, Jinyeop Song, **Minkyu Jeon** paper

Inf. Sci. 2023 — **Randomly shuffled convolution for self-supervised representation learning**
Youngjin Oh[*], **Minkyu Jeon**[*], Dohwan Ko, Hyunwoo J. Kim paper / code

NeurIPS 2022 — **SageMix: Saliency-guided mixup for point clouds**
Sanghyeok Lee[*], **Minkyu Jeon**[*], Injae Kim, Yunyang Xiong, Hyunwoo J. Kim paper / code

ECCV 2022 — **k-SALSA: k-anonymous synthetic averaging of retinal images via local style alignment**
Minkyu Jeon, Hyeonjin Park, Hyunwoo J. Kim, Michael Morley, Hyunghoon Cho paper / code

IEEE Access 2021 — **Learning to balance local losses via meta-learning**
Seungdong Yoa, **Minkyu Jeon**, Youngjin Oh, Hyunwoo J. Kim paper

TALKS

Dec 2025	NeurIPS 2025 MLSB Workshop (oral, with Jeffrey Gu): amortized heterogeneous cryo-EM reconstruction
Aug 2024	Flatiron Institute, Simons Foundation (invited): CryoBench
Feb 2024	CMU MetaMobility Lab (invited lecture): k-SALSA

HONORS

2024–2026	**Asan Foundation Biomedical Science Scholarship** (three-year doctoral scholarship)
2021	**I-Corps Program**, $35,000 government support for startup customer discovery
2019	Dean's List and 40% tuition scholarship, Konkuk University
2016	Certificate of Commendation, Republic of Korea Marine Corps (military service 2015–2016)

SERVICE & TEACHING

Reviewing	AAAI 2027 (program committee); NeurIPS 2025–26; ICLR 2025–27; CVPR 2026; ECCV 2026; NeurIPS workshops 2023–24; *Information Sciences* 2024
Princeton	TA, COS302 Mathematics for Numerical Computing and ML (Fall 2024); Instructor, AI4ALL (Summer 2024)
Korea Univ.	TA, Artificial Intelligence (2021) and AI Security (2020); Tutorial Teaching Fellow, Deep Learning (2020)

SKILLS

Machine learning	PyTorch, TensorFlow; diffusion and flow matching; transformers; implicit neural representations; LLM agents (vLLM)
Computing	Python, R, C; Linux; SLURM multi-GPU clusters; Git